Type MS · Completing the square

Answer 1 of 12 · Question 3 part a · 3 marks

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3 a	$8\left(x^2+\frac{10}{8}x\right)-3$	M1
	$8\left(x+\frac{5}{8}\right)^2-\frac{200}{64}-3=8\left(x+\frac{5}{8}\right)^2-\frac{49}{8}$	M1
	So $A=8$ $B=\frac{5}{8}$ $C=-\frac{49}{8}$	A1 (3)
b (i)	$x=-\frac{5}{8}$	B1 ft
(ii)	$f(x)_{min}=-\frac{49}{8}$	B1 ft (2)
c	$8\left(x+\frac{5}{8}\right)^2-\frac{49}{8}=0\Rightarrow x=-\frac{3}{2}$ $x=\frac{1}{4}$	M1 A1 (2)
d	$8x^2+10x-3=2x+13\Rightarrow 8x^2+8x-16=0$ oe	M1
	$x^2+x-2=0=(x+2)(x-1)\Rightarrow x=-2,\ x=1$	M1 A1
	$y=9$ $y=15$ Coordinates are $(-2,9)$ and $(1,15)$	A1 (4)
e	Correct curve or line Correct curve and line	B1 B1 (2)
		Total 13 marks

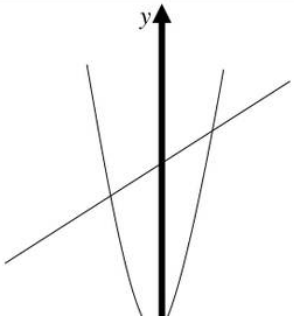
Type MS · Completing the square

Answer 1 of 12 · Question 3 part a · 3 marks

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(b) (i)	B1ft	For $x=-\frac{5}{8}$ ft their $-B$ Allow differentiation: $\frac{dy}{dx}=16x+10=0\Rightarrow x=-\frac{10}{16}$ which must be correct.
(ii)	B1ft	For $f(x)_{min}=-\frac{49}{8}$ ft their C If they differentiate, allow a ft from their differentiation.
	Incorrect or no labelling of parts. If the responses are labelled incorrectly, mark as labelled. If there is no labelling, treat the first as (i) and the second as (ii) and mark accordingly.	
(c)	M1	For setting the given f(x) = 0 and solving using any method. See General Guidance.
	A1	For $x=-\frac{3}{2}$ and $x=\frac{1}{4}$
(d)	M1	For setting $8x^2+10x-3=2x+13$ and forming a 3TQ which as a minimum must be $8x^2+8x\pm C$ where C is a constant, or any simplification, e.g., $x^2+x\pm K$ where K is a constant.
	M1	For attempting to solve their 3TQ. See General Guidance.
	A1	For **both** $x=-2$, $x=1$
	A1	For $(-2,9)$ and $(1,15)$ Accept $x=-2$, $y=9$ and $x=1$, $y=15$ paired correctly.
	ALT – solves the equation in y	
	M1	Substitutes $x=\frac{y-13}{2}$ into $y=8x^2+10x-3$ and forms a 3TQ which as a minimum must be of the form $2y^2-36y\pm X$ $X\neq 0$ $\Rightarrow y=8\left(\frac{y-13}{2}\right)^2+10\left(\frac{y-13}{2}\right)-3\Rightarrow 2y^2-36y+75=0$
	M1	For attempting to solve their 3TQ. See General Guidance.
	A1	For **both** $y=9$ and $y=15$
	A1	For $(-2,9)$ and $(1,15)$ Accept $x=-2$, $y=9$ and $x=1$, $y=15$ paired correctly.
(e)	B1	For correct curve (intersections with x-axis are at $(-1.5, 0)$ and $(0.25, 0)$) **or** line (Intersections are $(-6.5, 0)$ and $(0, 13)$ drawn. **We do not need to see any of these points marked. These are for guidance only.** The question asks for a sketch, accept a reasonable attempt. • Accept a positive quadratic curve with the minimum point below the x-axis, and one branch either side of the y-axis. • Accept a straight line with a positive gradient where the intersection with the y-axis is positive.
	B1	For correct curve **and** line drawn

SKETCH OF CURVE AND LINE



Type MS · Completing the square

Answer 1 of 12 · Question 3 part a · 3 marks

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Question number	Scheme	Marks
4	When $x=\frac{\pi}{2}$ $y=\frac{\pi^3}{8}$ So $\left(\frac{\pi}{2},\frac{\pi^3}{8}\right)$	B1
	$\frac{dy}{dx}=3x^2\sin x+x^3\cos x$	M1 A1 A1
	When $x=\frac{\pi}{2}$ $\frac{dy}{dx}=3\left(\frac{\pi}{2}\right)^2\sin\left(\frac{\pi}{2}\right)+\left(\frac{\pi}{2}\right)^3\cos\left(\frac{\pi}{2}\right)=\left[,\frac{3\pi^2}{4},\right]$	M1
	$y-\frac{\pi^3}{8}=\frac{3\pi^2}{4}\left(x-\frac{\pi}{2}\right)$	M1
	$y=\frac{3}{4}\pi^2x-\frac{1}{4}\pi^3$	A1
		Total 7 marks

Type MS · Completing the square

Answer 1 of 12 · Question 3 part a · 3 marks

2023 JAN · Paper 02 · extract 4 of 20 · find part a in this mark scheme extract

Mark	Notes
Note: In this question, all substitution of angle values must be in Radians only.	
B1	For obtaining $y = \frac{\pi^3}{8}$ [allow $\left(\frac{\pi}{2}\right)^3$ and also awrt $y = 3.88$]
M1	For an attempt to use the product rule. The definition of an attempt is as follows: • There must be a correct attempt to differentiate both terms. $\sin x \Rightarrow \cos x \quad x^3 \Rightarrow ax^2$ where $x \neq 0$ • The correct formula must be used. i.e., it must be a sum of their two terms.
A1	At least one term must be correct. Either $3x^2 \sin x$ or $x^3 \cos x$
A1	For $3x^2 \sin x + x^3 \cos x$ oe Ignore any subsequent simplification once you have seen the correct answer – even if the simplification is incorrect.
M1	For substitution of $\frac{\pi}{2}$ into their $\frac{dy}{dx} = \left['\frac{3}{4}\pi^2'\right]$ provided it is a changed expression. Allow a value of awrt 7.4(0) **NOTE:** You must see a full substitution of $\frac{\pi}{2}$ into their $\frac{dy}{dx}$ if their expression for $\frac{dy}{dx}$ is incorrect.
M1	For a correct method for finding the equation of a line using their value of y, dy/dx and the given x [allow $x = 1.57\ldots$]. This must be applied correctly. Either uses the formula with their values, or if uses $y = mx + c$ they must reach a value for c before this mark can be awarded. Do **not** allow processing errors to find the value of c
A1	For $y = \frac{3}{4}\pi^2 x - \frac{1}{4}\pi^3$ Allow $y = 7.4(0)x - 7.75$ or better values. Do not allow a mixture of decimals and exact values.

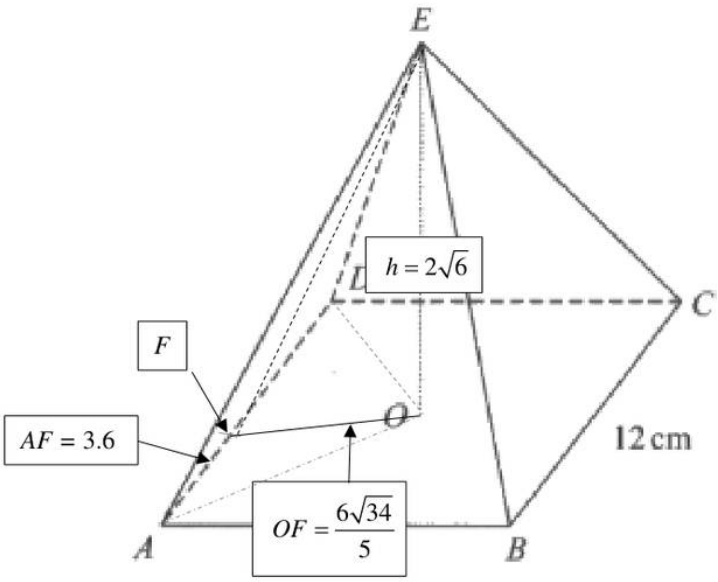
Type MS · Completing the square

Answer 1 of 12 · Question 3 part a · 3 marks

2023 JAN · Paper 02 · extract 5 of 20 · find part a in this mark scheme extract

Question number	Scheme	Marks
5 a	$OC = \frac{1}{2}\sqrt{12^2+12^2} = \left[6\sqrt{2}\right]$ or $AC = \sqrt{12^2+12^2} = \left[12\sqrt{2}\right]$	M1
	$h = EO = 6\sqrt{2} \times \frac{\sqrt{3}}{3} = 2\sqrt{6}$ oe	M1 A1
		(3)
b	Midpoint of AD to $F = \left(6 - \frac{12}{1+4}\right)$ [= 3.6]	M1
	$OF = \sqrt{6^2+3.6^2} = \frac{6\sqrt{34}}{5}$	M1
	$\tan\theta = \frac{2\sqrt{6}}{\frac{6\sqrt{34}}{5}} = 35^\circ$	M1 A1
		(4)
		Total 7 marks

USEFUL SKETCH



Type MS · Completing the square

Answer 1 of 12 · Question 3 part a · 3 marks

2023 JAN · Paper 02 · extract 6 of 20 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For using a correct Pythagoras theorem to find either OC, OA $\left(6\sqrt{2}\right)$ or AC $\left(12\sqrt{2}\right)$
	M1	For using the correct trigonometry (tan ratio): $h = '6\sqrt{2}'\tan 30^\circ = \left[2\sqrt{6}\right]$ or equivalent. Eg $\tan 30 = \dfrac{h}{6\sqrt{2}} \Rightarrow h = ...$
	A1	For $2\sqrt{6}$ oe
(b)	M1	For correct expression for the midpoint of AD to F $\left(6 - \dfrac{12}{1+4}\right)$ or 3.6 oe seen
	M1	For the correct use of Pythagoras' to find OF $\dfrac{6\sqrt{34}}{5}$ oe [6.997..] ft their 3.6
	ALT	
	M1	Finds the length OF using cosine rule. $OF = \sqrt{2.4^2 + \left(6\sqrt{2}\right)^2 - 2\times 2.4\times 6\sqrt{2}\times\cos 45} = \left[\dfrac{6\sqrt{34}}{5}\right]$ OR $OF = \sqrt{9.6^2 + \left(6\sqrt{2}\right)^2 - 2\times 9.6\times 6\sqrt{2}\times\cos 45} = \left[\dfrac{6\sqrt{34}}{5}\right]$ OR $\cos 45^\circ = \dfrac{9.6^2 + \left(6\sqrt{2}\right)^2 - OF^2}{2\times 9.6\times 6\sqrt{2}} \Rightarrow OF = ...$ oe [6.997..]
	M1	For the correct evaluation of their cosine rule. $\left(\dfrac{6\sqrt{34}}{5}\text{ oe}\right)$ Accept awrt 7.00 [6.997142…]
	dM1	For the correct use of $\tan\theta = \dfrac{'EO'}{'OF'}$ This mark is dependent on both previous M marks. They must have a valid method to find OF for the award of this mark.
	A1	For 35(°) or better (Calculator value is 34.997…°.)
	Note: There are other methods – if unsure, send to Review.	

Type MS · Completing the square

Answer 1 of 12 · Question 3 part a · 3 marks

2023 JAN · Paper 02 · extract 7 of 20 · find part a in this mark scheme extract

Question number	Scheme	Marks
6 a	$r=\frac{q(2p-3)}{q(2p+3)}=\frac{q(2p+3)}{q(4p+1)}$	M1
	$(4p+1)(2p-3)=(2p+3)^2$	dM1
	$2p^2-11p-6=0$	A1
	$(2p+1)(p-6)=0$	M1
	$p=-\frac{1}{2}$ or $p=6$	A1 (5)
b	When $p=6$ $r=\frac{3}{5}$ and $U_1=q\left(4\times'6'+1\right)=\left(25q\right)$	M1
	$S_\infty=\frac{25q}{\frac{2}{5}}=250\Rightarrow q=4$	dM1 A1 (3)
		Total 8 marks

Type MS · Completing the square

Answer 1 of 12 · Question 3 part a · 3 marks

2023 JAN · Paper 02 · extract 8 of 20 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For $r = \frac{\text{3rd term}}{\text{2nd term}} = \frac{\text{2nd term}}{\text{1st term}}$
	dM1	For attempting to remove the denominators/simplifying the expression to attempt to obtain a 3TQ. Allow one processing error. Allow a minimally acceptable $4p^2 \pm Xp \pm Y = (0)$ or $2p^2 \pm Pp \pm Q = (0)$ $X, Y, P, Q \neq 0$ **Note:** This mark is dependent on the previous M mark.
	A1	For obtaining $2p^2 - 11p - 6 = (0)$ or equivalent. Their working will give them $4p^2 - 22p - 12 = (0)$ before further simplification
	M1	For a valid attempt to solve the 3TQ. Note this is an independent mark for their 3TQ. See General Guidance. The 3TQ must have come from some attempted manipulation involving $q(2p-3)$, $q(2p+3)$ and $q(4p+1)$
	A1	For $p = -\frac{1}{2}$ or $p = 6$
(b)	M1	For finding: • A value of r for their p. Allow any value of r $r \neq 0$ E.g. $r = \frac{(2\times'6'-3)}{(2\times'6'+3)} = ...$ $\left(\text{Note: for } p = -\frac{1}{2} \quad r = -2\right)$ • A value for U_1 $U_1 = q(4\times'6'+1) = ...$ $\left(\text{Note: for } p = -\frac{1}{2}, U_1 = -q\right)$ You may see a = 100 after later working to find the sum to infinity. If their values of p are incorrect, we **must** see working here. **NOTE:** This is a B mark in Epen.
	dM1	For use of $S_\infty = \frac{a}{1-r} = 250$ with their r and their U_1 provided $\lvert r\rvert < 1$ If they use the formula for the sum to infinity on an $\lvert r\rvert > 1$ withhold this mark even if they use it on one valid and one invalid attempt. **Note:** This mark is dependent on the previous M mark.
	A1	For $q = 4$

Type MS · Completing the square

Answer 1 of 12 · Question 3 part a · 3 marks

2023 JAN · Paper 02 · extract 9 of 20 · find part a in this mark scheme extract

Question number	Scheme	Marks
7 a	$\frac{dy}{dx} = 2e^{2x}\cos 2x - 2e^{2x}\sin 2x$	M1 A1 A1
	$\frac{dy}{dx} = 2y - 2e^{2x}\sin 2x$ *	A1 cso (4)
b	$\frac{d^2y}{dx^2} = 2\frac{dy}{dx} - 4e^{2x}\sin 2x - 4e^{2x}\cos 2x$	M1 A1
	$\frac{d^2y}{dx^2} = 2\frac{dy}{dx} + 2\left(\frac{dy}{dx} - 2y\right) - 4y$	dM1ddM1
	$\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y$ *	A1 cso (5)
		Total 9 marks

Part	Mark	Notes
(a)	M1	For an attempt to use the product rule. The definition of an attempt is as follows: • There must be a correct attempt to differentiate both terms. $\cos 2x \Rightarrow \pm 2\sin 2x \quad e^{2x} \Rightarrow 2e^{2x}$ where $x \neq 0$ • The terms must be added The correct formula must be used.
	A1	For one term correct
	A1	For both terms correct
	A1 cso	For obtaining the given result with no errors seen.
(b)	**Method 1**	
	M1	For attempting to differentiate $\frac{dy}{dx}$ to obtain $\frac{d^2y}{dx^2}$ $\frac{d^2y}{dx^2} = 4e^{2x}\cos 2x - 4e^{2x}\sin 2x - 4e^{2x}\sin 2x - 4e^{2x}\cos 2x$ For this mark accept either: $\left(4e^{2x}\cos 2x - 4e^{2x}\sin 2x\right)$ **OR** $-4e^{2x}\sin 2x - 4e^{2x}\cos 2x$
	A1	For the correct $\frac{d^2y}{dx^2}$ (This is an M mark in Epen)
	dM1	For simplifying $\frac{d^2y}{dx^2}$ to obtain $\frac{d^2y}{dx^2} = -8e^{2x}\sin 2x$ (A mark in Epen) **This mark is dependent on the first M mark**

Type MS · Completing the square

Answer 1 of 12 · Question 3 part a · 3 marks

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	ddM1	For using the substitution $-2e^{2x}\sin 2x = \frac{dy}{dx} - 2y \Rightarrow -8e^{2x}\sin 2x = 4\frac{dy}{dx} - 8y$ **This mark is dependent on BOTH previous M marks.**
	A1cso	For obtaining the given result with no errors seen. $\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y$
	Method 2	
	M1	For attempting to differentiate $\frac{dy}{dx}$ to obtain $\frac{d^2y}{dx^2}$ $\frac{d^2y}{dx^2} = 2\frac{dy}{dx} - 4e^{2x}\sin 2x - 4e^{2x}\cos 2x$ For this mark accept either: $2\frac{dy}{dx}$ or $-4e^{2x}\sin 2x - 4e^{2x}\cos 2x$
	A1	For the correct $\frac{d^2y}{dx^2}$ **This is an M mark in Epen**
	dM1	For substituting $-4e^{2x}\cos 2x \Rightarrow -2y$ **This is an A mark in Epen**
	ddM1	For using the substitution $-4e^{2x}\sin 2x = 2\frac{dy}{dx} - 4y$
	A1	For obtaining the given result with no errors seen. $\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y$
	Method 3 – Works form LHS and RHS together.	
	M1	For attempting to differentiate $\frac{dy}{dx}$ to obtain $\frac{d^2y}{dx^2}$ $\frac{d^2y}{dx^2} = 4e^{2x}\cos 2x - 4e^{2x}\sin 2x - 4e^{2x}\sin 2x - 4e^{2x}\cos 2x$ For this mark accept either: $\left(4e^{2x}\cos 2x - 4e^{2x}\sin 2x\right)$ **OR** $-4e^{2x}\sin 2x - 4e^{2x}\cos 2x$
	A1	For the correct $\frac{d^2y}{dx^2}$ **This is an M mark in Epen**
	dM1	For simplifying $\frac{d^2y}{dx^2}$ to obtain $\frac{d^2y}{dx^2} = -8e^{2x}\sin 2x$ (A mark in Epen) **This mark is dependent on the first M mark**
	ddM1	Multiplies out the given expression in (b) $\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y = 4\left(2e^{2x}\cos 2x - 2e^{2x}\sin 2x\right) - 8\left(e^{2x}\cos 2x\right)$ $= 8e^{2x}\cos 2x - 8e^{2x}\sin 2x - 8e^{2x}\cos 2x$ $= -8e^{2x}\sin 2x$

Type MS · Completing the square

Answer 1 of 12 · Question 3 part a · 3 marks

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		This mark is dependent on BOTH previous M marks.
	A1	For a conclusion. A simple # sign, 'shown', 'QED' or underlining is sufficient.

Question number	Scheme	Marks
8	$\alpha+\beta=4k\sqrt{2}$ and $\alpha\beta=2k^4-1$	B1 B1
	$(\alpha+\beta)^2=\alpha^2+\beta^2+2\alpha\beta\Rightarrow\left(4k\sqrt{2}\right)^2=66+2\left(2k^4-1\right)$	M1 A1
	$k^4-8k^2+16=0$	M1
	$\left(k^2-4\right)^2=0\Rightarrow k=2$	M1 A1
	$(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3$ **or** $\alpha^3+\beta^3=(\alpha+\beta)\left(\alpha^2+\beta^2-\alpha\beta\right)$	M1
	$\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$ **or** $\alpha^3+\beta^3=(\alpha+\beta)(66-\alpha\beta)$	A1
	$\alpha^3+\beta^3=\left(8\sqrt{2}\right)^3-3\times31\times8\sqrt{2}=280\sqrt{2}$ **or** $8\sqrt{2}(66-31)=280\sqrt{2}$	M1
	p=280	A1 (11)
		Total 11 marks

Mark	Notes
B1	For **either** $\alpha+\beta=4k\sqrt{2}$ **or** $\alpha\beta=2k^4-1$
B1	For **both** $\alpha+\beta=4k\sqrt{2}$ **and** $\alpha\beta=2k^4-1$
M1	For the correct algebra on $\alpha^2+\beta^2$ (in any order) **and** substitution of their values of $\alpha\beta$ and $\alpha+\beta$ providing both sum and product are in terms of k. $(\alpha+\beta)^2=\alpha^2+\beta^2+2\alpha\beta\Rightarrow\alpha^2+\beta^2=\left('4k\sqrt{2}'\right)^2-2\left('2k^4-1'\right)$
A1	For obtaining $\left(4k\sqrt{2}\right)^2=66+2\left(2k^4-1\right)$ in any order.
M1	For simplifying to form a 3TQ in k^4 i.e., $4k^4-32k^2+64=0$ oe Accept as a minimum $4k^4-32k^2\pm Q=(0)\quad Q\neq0$
M1	For factorising or solving the 3TQ using any valid method. See General Guidance.
A1	For $k=2$ If they also give $k=-2$ withhold this mark.

Type MS · Completing the square

Answer 1 of 12 · Question 3 part a · 3 marks

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Method 1	
M1	For expanding $(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3$ or $\alpha^3+\beta^3=(\alpha+\beta)(\alpha^2+\beta^2-\alpha\beta)$
A1	For obtaining $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$ or $\alpha^3+\beta^3=(\alpha+\beta)(66-\alpha\beta)$ This must be such that $\alpha\beta$ and $\alpha+\beta$ can be substituted in directly.
M1	For substitution of $\alpha+\beta$ and $\alpha\beta$ for their positive value of k into a **correct** expansion of $\alpha^3+\beta^3$ **NOTE:** If they do not obtain $k=2$, then full substitution of their numerical value for k into $\alpha+\beta$ **and** $\alpha\beta$ must be seen for the award of this mark. For example: $\alpha^3+\beta^3=\left(4\times\text{'their } k\text{'}\sqrt{2}\right)^3-3\times\left(2\times\text{'their } k^4\text{'}-1\right)\times\left(4\times\text{'their } k\text{'}\sqrt{2}\right)$
A1	For $p=280$
Method 2	
M1	Finds the exact value of α and β Solves the equations $\left(\alpha\beta=2k^4-1 \text{ and } \alpha+\beta=4k\sqrt{2}\right)$ simultaneously to give a value for α and β $\alpha=\frac{31}{\beta}\Rightarrow\alpha+\beta=\frac{31}{\beta}+\beta=8\sqrt{2}\Rightarrow\beta^2-8\sqrt{2}\beta+31=0$ $\Rightarrow\beta=\ldots\quad\alpha=\ldots$
A1	For $\alpha=1+4\sqrt{2}\quad\beta=-1+4\sqrt{2}$ OR $\beta=1+4\sqrt{2}\quad\alpha=-1+4\sqrt{2}$
M1	Substitutes these values into $\alpha^3+\beta^3=p\sqrt{2}$ to find a value for p $\left(1+4\sqrt{2}\right)^3+\left(-1+4\sqrt{2}\right)^3=24\sqrt{2}+256\sqrt{2}=280\sqrt{2}\Rightarrow p=\ldots$
A1	$p=280$

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Answer 1 of 12 · Question 3 part a · 3 marks

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Question number	Scheme	Marks
9		
	$\frac{dA}{dt} = 0.45$	B1
	$V = x^3 \Rightarrow \frac{dV}{dx} = 3x^2$	B1
	$A = 6x^2 \Rightarrow \frac{dA}{dx} = 12x$	B1
	$384 = 6x^2 \Rightarrow x = 8$	M1
	$\frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dA} \times \frac{dA}{dt}$	M1
	$\frac{dV}{dt} = 3x^2 \times \frac{1}{12x} \times 0.45 \left[= \frac{9x}{80} \right]$ oe	dM1
	When $x = 8$ $\frac{dV}{dt} = 0.9\,\text{cm}^3/\text{s}$	A1
		Total 7 marks

Type MS · Completing the square

Answer 1 of 12 · Question 3 part a · 3 marks

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Mark	Notes
B1	For $\frac{dA}{dt} = 0.45$ seen anywhere in their working Accept other letters, for example S for the area $\frac{dS}{dt} = 0.45$
B1	For $\frac{dV}{dx} = 3x^2$ Accept also other letters in place of x such as r for example.
B1	For $\frac{dA}{dx} = 12x$ Accept also other letters in place of x such as r for example.
M1	For setting $384 = 6x^2$ and proceeding to a correct method leading to a value of x Award this mark when they obtain $x^2 = 64 \Rightarrow x = ...$
M1	For a correct expression of the chain rule seen or implied. i.e., $\frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dA} \times \frac{dA}{dt}$ They may complete this in two stages. So you may see for example: $\frac{dx}{dt} = \frac{1}{\frac{dA}{dx}} \times \frac{dA}{dt}$ **AND** $\frac{dV}{dt} = \frac{dx}{dt} \times \frac{dV}{dx}$
dM1	For substituting their values into a **correct** chain rule. $\frac{dV}{dt} = 3(8)^2 \times \frac{1}{12(8)} \times 0.45$ This mark is dependent on the previous M mark scored.
A1	For 0.9 (cm^3/s)

Type MS · Completing the square

Answer 1 of 12 · Question 3 part a · 3 marks

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Question number	Scheme	Marks
10 a (i)	$\sin 2\theta = \sin\theta\cos\theta + \cos\theta\sin\theta = 2\sin\theta\cos\theta$ *	M1 A1 cso (2)
a (ii)	$\cos 2\theta = \cos^2\theta - \sin^2\theta = \cos^2\theta - (1-\cos^2\theta)$	M1 M1
	$= 2\cos^2\theta - 1$ *	A1 cso (3)
b	$\sin 2\theta - \tan\theta = 2\sin\theta\cos\theta - \dfrac{\sin\theta}{\cos\theta}$	M1
	$\sin\theta\left(2\cos\theta - \dfrac{1}{\cos\theta}\right)$	M1
	$\sin\theta\left(\dfrac{2\cos^2\theta - 1}{\cos\theta}\right) = \tan\theta\cos 2\theta$ *	dM1 A1cso (4)
c	$\tan x = 0$ so $x = 180$	B1
	$\cos 2x = 0$ so $2x = 90, 270, 450, 630$	M1
	$x = 45, 135, 225, 315$	A1 A1 (4)
		Total 13 marks

Part	Mark	Notes
(a)(i)	M1	For the correct use of: $\sin(A+B) = \sin A\cos B + \cos A\sin B \Rightarrow \sin 2\theta = \sin\theta\cos\theta + \sin\theta\cos\theta$ Allow any letter to be used for this mark.
	A1 Cso	For obtaining the given result $\sin 2\theta = 2\sin\theta\cos\theta$ or allow $\sin(\theta+\theta) = 2\sin\theta\cos\theta$
(a)(ii)	M1	For the correct use of: $\cos(A+B) = \cos A\cos B - \sin A\sin B$ $\Rightarrow \cos 2A = \cos^2 A - \sin^2 A$ or $\cos(A+A) = \cos A\cos A - \sin A\sin A$ Allow any letter to be used for this mark.
	M1	For use of $\sin^2 A + \cos^2 A = 1$ and substituting Allow any letter to be used for this mark.
	A1 cso	For obtaining the given result $\cos 2\theta = 2\cos^2\theta - 1$ or allow $\cos(\theta+\theta) = 2\cos^2\theta - 1$
(b)	M1	For correct use of $\sin 2\theta = 2\sin\theta\cos\theta$ **and** $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$

Type MS · Completing the square

Answer 1 of 12 · Question 3 part a · 3 marks

2023 JAN · Paper 02 · extract 16 of 20 · find part a in this mark scheme extract

	M1	For factorising the resulting expression to give $\sin\theta\left(2\cos\theta-\frac{1}{\cos\theta}\right)$ **and** using a common denominator of $\cos\theta$ correctly to obtain $\sin\theta\left(\frac{2\cos^2\theta-1}{\cos\theta}\right)$ oe e.g. $\frac{\sin\theta}{\cos\theta}\left(2\cos^2\theta-1\right)$
	dM1	For use of $2\cos^2\theta-1=\cos 2\theta$ **Note:** This mark is dependent on both previous M marks
	A1 cso	For obtaining the given result
	ALT - Allow candidates to work from both sides. For example.	
	M1	For correct use of $\sin 2\theta=2\sin\theta\cos\theta$ **and** $\tan\theta=\frac{\sin\theta}{\cos\theta}$ $\sin 2\theta-\tan\theta=\tan\theta\cos 2\theta$ $\Rightarrow 2\sin\theta\cos\theta-\frac{\sin\theta}{\cos\theta}=\frac{\sin\theta}{\cos\theta}\left(2\cos^2\theta-1\right)$
	M1	For using a common denominator on the LHS to give: $\frac{2\sin\theta\cos^2\theta-\sin\theta}{\cos\theta}=\frac{\sin\theta\left(2\cos^2\theta-1\right)}{\cos\theta}$
	dM1	Multiplies or factorises one side to make LHS = RHS For example; $\frac{2\sin\theta\left(\cos^2\theta-1\right)}{\cos\theta}=\frac{2\sin\theta\left(\cos^2\theta-1\right)}{\cos\theta}$
	A1 cso	Concludes RHS = LHS therefore shown. Note there must be no errors seen. If unsure, send to Review.
(c)	Some candidates are starting again without using the given expression in (b) and are getting their angles from cos x rather than cos $2x$ Allow the use of θ or any other variable throughout this part of the question. As a general principle of marking this part: • B1 – For only 180° seen. If 0° is included, then B0. • M1 – For either $2x=90,(270,450,630)$ **OR** $x=45^\circ$ or 135° or 225° or 315° or any other correct angle in or out of range. • A1 – For any two correct angles apart from 180° • A1 – For all four correct angles, apart form 180°	
	B1	For $x=180$ If $x=0$ included then B0
	M1	For obtaining $2x=90,(270,450,630)$
	A1	For any two correct angles from 45°, 135°, 225° or 315°
	A1	For all four angles from $\cos x$ or $\cos 2x$
	For extra angles within range, penalise the last A mark only. For example; If they have [coming from a correct method] $x=45°, 67.5°, 135°, 225°, 315°$ score M1A1A0 If they have $x=45°, 67.5°, 135°$ then award M1A1A0	

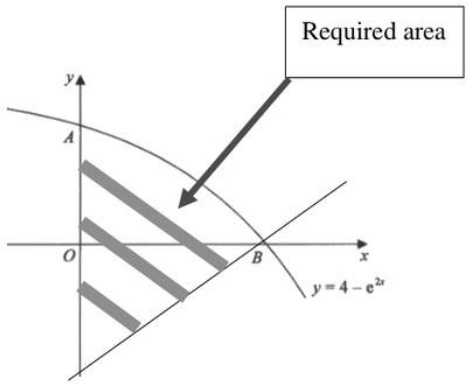
Type MS · Completing the square

Answer 1 of 12 · Question 3 part a · 3 marks

2023 JAN · Paper 02 · extract 17 of 20 · find part a in this mark scheme extract

Question number	Scheme	Marks
11 a	When $x=0 \quad y=3$	B1
	When $y=0 \quad 0=4-e^{2x} \Rightarrow \ln 4 = 2x \Rightarrow x = \ln 2$	M1 A1 (3)
b	$\frac{dy}{dx} = -2e^{2x}$	M1
	When $x=\ln 2 \quad \frac{dy}{dx} = -8$	M1
	$y-0=\frac{1}{8}(x-\ln 2) \Rightarrow y=\frac{1}{8}x-\frac{1}{8}\ln 2$	M1 A1 (4)
c	$\int_0^{\ln 2}\left(4-e^{2x}\right)dx=\left[4x-\frac{1}{2}e^{2x}\right]_0^{\ln 2}=\left(4\ln 2-\frac{1}{2}e^{2\ln 2}\right)-\left(0-\frac{1}{2}e^{0}\right)$	M1M1M1
	$\frac{1}{2}(\ln 2)\left(\frac{1}{8}\ln 2\right)=\frac{1}{16}(\ln 2)^2$	M1M1
	$\frac{1}{8}\int_0^{\ln 2}\lvert x-\ln 2\rvert\,dx=\frac{1}{8}\left\lvert\frac{x^2}{2}-x\ln 2\right\rvert_0^{\ln 2}=\left\lvert\frac{(\ln 2)^2}{16}-\frac{(\ln 2)^2}{8}\right\rvert=\frac{(\ln 2)^2}{16}$	[M1M1]
	$4\ln 2-\frac{3}{2}+\frac{1}{16}(\ln 2)^2=1.3$	M1 A1 (7)
		Total 14 marks

USEFUL SKETCH



Part	Mark	Notes

Type MS · Completing the square

Answer 1 of 12 · Question 3 part a · 3 marks

2023 JAN · Paper 02 · extract 18 of 20 · find part a in this mark scheme extract

(a)(i) (ii)	B1	For $y = 3$
	M1	For $\ln 4 = 2x$ or $\sqrt{4} = \sqrt{e^{2x}}$ seen explicitly
	A1	For $x = \ln 2$
(b)	M1	For differentiating the given expression. This must be correct for this mark. $\frac{dy}{dx} = -2e^{2x}$
	M1	For substituting ln2 into their $\frac{dy}{dx} = -2e^{2\ln 2} = [-8]$
	M1	For a correct method for finding the equation of a straight line using their numerical perpendicular gradient and $y = 0$ and $x = \ln 2$ Award when they substitute into a correct formula, or if they use $y = mx + c$ award when c is obtained. (accept a decimal value for c for this mark awrt $c = -0.087$)
	A1	For $y = \frac{1}{8}x - \frac{1}{8}\ln 2$ or $y = \frac{x}{8} - \frac{\ln 2}{8}$ in exact form only.
(c)	**Area under curve**	
	M1	For a correct statement for the area under the curve with correct limits. Accept the limits either way around. Ignore poor notation. This mark can be implied by later correct work. $\int_0^{\ln 2}\left(4 - e^{2x}\right)dx$
	M1	For a minimally acceptable attempt to integrate as follows. $\int_0^{\ln 2}\left(4 - e^{2x}\right)dx = \left[4x \pm \frac{e^{2x}}{2}\right]$
	M1	For substitution of both limits into their integral of the curve. $A = \left(4\ln 2 - \frac{1}{2}e^{2\ln 2}\right) - \left(0 - \frac{1}{2}e^{0}\right) = \left[4\ln 2 - \frac{3}{2}\right] = [1.27258....]$
	Area of the triangle	
	M1	**Method 1** For a statement of the area. $A = \frac{1}{2}(\ln 2)\left('\frac{1}{8}\ln 2'\right)$ **Method 2** For a statement of the area, with limits either way around. $A = \frac{1}{8}\int_0^{\ln 2}(x - \ln 2)\,dx$ ft their equation of the line.
	M1	For a correct method to evaluate the area of the triangle. **Method 1** $A = \frac{1}{2}(\ln 2)\left('\frac{1}{8}\ln 2'\right) = \frac{1}{16}(\ln 2)^2 = [0.03002.....]$ **Method 2** The integration and substitution must be correct for this mark $A = \frac{1}{8}\int_0^{\ln 2}\lvert x - \ln 2\rvert\,dx = \frac{1}{8}\left\lvert\frac{x^2}{2} - x\ln 2\right\rvert_0^{\ln 2} = \left\lvert\frac{(\ln 2)^2}{16} - \frac{(\ln 2)^2}{8}\right\rvert = \frac{(\ln 2)^2}{16}$

Type MS · Completing the square

Answer 1 of 12 · Question 3 part a · 3 marks

2023 JAN · Paper 02 · extract 19 of 20 · find part a in this mark scheme extract

	Combined area	
	M1	For the total final area using their values $4\ln 2-\frac{3}{2}+\frac{1}{16}(\ln 2)^2$ **This is an A mark in Epen.**
	A1	For awrt 1.3 Also accept the exact Area = $4\ln 2-\frac{3}{2}+\frac{1}{16}(\ln 2)^2$

Type MS · Completing the square

Answer 1 of 12 · Question 3 part a · 3 marks

2023 JAN · Paper 02 · extract 20 of 20 · find part a in this mark scheme extract

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Type MS · Completing the square

Answer 2 of 12 · Question 10 part a · 3 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

Number		
10 (a)	$6x-x^2=-\left(x^2-6x\right)$ $-\left(x^2-6x\right)=-\left\{(x-3)^2-9\right\}\Rightarrow \mathrm{f}(x)=-(x-3)^2+9$ $D=-1,\ E=-3$ and $F=9$	M1A1A1 [3]
(b)	(i) $\mathrm{f}(x)_{\max}=9$ (ii) $x=3$	B1ft B1ft [2]
(c)	$6x-x^2=x^2-4x+8\Rightarrow 2x^2-10x+8=0$ $2x^2-10x+8=(2x-2)(x-4)\Rightarrow x=1,\ x=4$ $y=5,\ y=8$ Coordinates are (1, 5) and (4, 8)	M1 M1A1 A1 [4]
(d)	$\text{Area}=\int_1^4\left(6x-x^2\right)\mathrm{d}x-\int_1^4\left(x^2-4x+8\right)\mathrm{d}x=\int_1^4\left[-2x^2+10x-8\right]\mathrm{d}x$ $=\left[\frac{-2x^3}{3}+\frac{10x^2}{2}-8x\right]_1^4$ $=\left(\frac{-2\times4^3}{3}+\frac{10\times4^2}{2}-8\times4\right)-\left(\frac{-2\times1^3}{3}+\frac{10\times1^2}{2}-8\times1\right)=9\ (\text{units}^2)$	M1 M1 M1A1 [4]
		Total 13 marks

Type MS · Completing the square

Answer 2 of 12 · Question 10 part a · 3 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Completing the square

Answer 3 of 12 · Question 7 part a · 2 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

7(a)	$x^2-9x+14=\left(x-\frac{9}{2}\right)^2+14-\frac{81}{4}=\left(x-\frac{9}{2}\right)^2-\frac{25}{4}$	M1
	$a=-\frac{9}{2},\ b=-\frac{25}{4}$ oe	A1 (2)
(b)	(i) least value of $f(x)=-\frac{25}{4}$	B1ft
	(ii) least value when $x=\frac{9}{2}$	B1ft (2)
(c)	$x+5=x^2-9x+14$	M1
	$x^2-10x+9=0\Rightarrow(x-9)(x-1)=0$	M1
	Points are $(9,14)$ $(1,6)$	A1A1 (4)
(d)	Area $\int_1^9\left((x+5)-\left(x^2-9x+14\right)\right)dx=\int_1^9\left(-x^2+10x-9\right)dx$	M1
	$=\left[-\frac{x^3}{3}+5x^2-9x\right]_1^9$	M1A1
	$=(-243+405-81)-\left(-\frac{1}{3}+5-9\right)=85\frac{1}{3}$	M1A1 (5)
		[13]

Part	Mark	Notes
(a)	M1	For attempting to complete the square to achieve as a minimum $x^2-9x+14=\left(x\pm\frac{9}{2}\right)^2+14-k$ where k is a constant
	A1	For the correct expression $x^2-9x+14=\left(x-\frac{9}{2}\right)^2-\frac{25}{4}$ or $a=-\frac{9}{2},\ b=-\frac{25}{4}$ oe stated
(b)(i)	B1ft	For the correct value $f(x)=-\frac{25}{4}$ follow through their value of '$-\frac{25}{4}$'
(ii)	B1ft	For the correct value of $x=\frac{9}{2}$ provided they have $\left(x-\frac{9}{2}\right)^2$ …in part (a). Follow through their value of '$\frac{9}{2}$'

Type MS · Completing the square

Answer 3 of 12 · Question 7 part a · 2 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

(c)	M1	For equating the equation of the line with the equation of C $x+5=x^2-9x+14\Rightarrow x^2-10x+9=0$ **and** attempting to form a 3TQ
	M1	Attempts to solve their 3TQ by any method, provided it is the result of equating the line with C $x^2-10x+9=0\Rightarrow(x-9)(x-1)=0$
	A1	For the correct coordinates of **either** $(9,14)$ or $(1,6)$
	A1	For both correct pairs of coordinates $(9,14)$ and $(1,6)$
(d)	M1	For a correct expression for the required area with both limits correct. (ft their limits from (c)) Award this mark if they have 'curve – line' but otherwise correct. $\int_1^9\left((x+5)-\left(x^2-9x+14\right)\right)\mathrm{d}x=\left[\int_1^9\left(-x^2+10x-9\right)\mathrm{d}x\right]$, accept $\int_1^9\left(x^2-10x+9\right)\mathrm{d}x$ **OR** Area under the trapezium – curve $\frac{1}{2}\times8\times(6+14)-\int_1^9\left(x^2-9x+14\right)\mathrm{d}x$
	M1	For attempting to integrate the equation for the combined expression or the curve only.
	A1	For the correct integrated expression for required area. Ignore limits for this mark – even if they are absent altogether. Area = $=\left[-\frac{x^3}{3}+5x^2-9x\right]_1^9$ accept $\left[\frac{x^3}{3}-5x^2+9x\right]_1^9$ **OR** $\frac{1}{2}\times8\times(6+14)-\left(\frac{x^3}{3}-\frac{9x^2}{2}+14x\right)_1^9$ **OR** $\left(\frac{x^2}{2}+5x\right)_1^9-\left(\frac{x^3}{3}-\frac{9x^2}{2}+14x\right)_1^9$ or $\left(\frac{x^3}{3}-\frac{9x^2}{2}+14x\right)_1^9-\left(\frac{x^2}{2}+5x\right)_1^9$
	M1	For substituting their limits (x – coordinates from part(c)) into their integrated expression. $=(-243+405-81)-\left(-\frac{1}{3}+5-9\right)=\ldots$ **OR** $80-\left[\left(\frac{9^3}{3}-\frac{9\times9^2}{2}+14\times9\right)-\left(\frac{1^3}{3}-\frac{9\times1^2}{2}+14\times1\right)\right]=\ldots$ **OR** $\left[\left(\frac{9^2}{2}+5\times9\right)-\left(\frac{1^2}{2}+5\times1\right)\right]-\left[\left(\frac{9^3}{3}-\frac{9\times9^2}{2}+14\times9\right)-\left(\frac{1^3}{3}-\frac{9\times1^2}{2}+14\times1\right)\right]=\ldots$
	A1	For the correct area of $85\frac{1}{3}$ or $\frac{256}{3}$ If they get a value of $-85\frac{1}{3}$ they must give a **final** value of $85\frac{1}{3}$ for this mark.

Type MS · Completing the square

Answer 3 of 12 · Question 7 part a · 2 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question Number	Scheme	Marks
	e^x	

Type MS · Completing the square

Answer 4 of 12 · Question 2 part a · 3 marks

2022 JAN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

2 (a)	$f(x)=2\left(x^2-6x\right)+5$ OR $f(x)=2\left(x^2-6x+\frac{5}{2}\right)$	M1
	$f(x)=2\left(\left[x-3\right]^2-k\right)+5$ OR $f(x)=2\left(\left[x-3\right]^2-k+\frac{5}{2}\right)$	M1
	$f(x)=2\left[x-3\right]^2-13$	A1
	$a=2,\ b=-3,\ c=-13$ Allow values embedded in f (x)	[3]
ALT – Equating coefficients		
	$ax^2+2abx+\left(ab^2+c\right)\equiv 2x^2-12x+5\Rightarrow a=2$	[M1
	$2ab=-12\Rightarrow 2\times 2\times b=-12\Rightarrow b=-3$	M1
	$ab^2+c=5\Rightarrow 2\times(-3)^2+c=5\Rightarrow c=5-18=-13$	A1]
(b)	$2\left[x-'3'\right]^2-'13'>37\Rightarrow 2\left[x-'3'\right]^2=50\Rightarrow\left[x-3\right]^2=25\Rightarrow x='3'\pm\ldots$	M1
	Critical values $x=-2,8$	A1
	$\{x<-2\}\cup\{x>8\}$	M1A1 [4]
ALT - solving inequality without use of completed square form		
	$2x^2-12x-32>0\ \Rightarrow\ x^2-6x-16>0\ \Rightarrow\ (x+2)(x-8)>0$	[M1
	Critical values $x=-2,8$	A1
	$\{x<-2\}\cup\{x>8\}$	M1A1]
		Total 7 marks

Type MS · Completing the square

Answer 4 of 12 · Question 2 part a · 3 marks

2022 JAN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

		$ab^2+c=5 \Rightarrow 2\times(-3)^2+c=5 \Rightarrow c=5-18=-13$
	A1	For correct values of a, b and c. $a=2,\ b=-3,\ c=-13$ Allow embedded answers i.e. $f(x)=2[x-3]^2-13$
(b)	**M1**	Sets f(x) > 37 and uses their result from part (a) [provided it is in the form $f(x)=2(x\pm P)^2\pm Q$] and attempts to find **two** critical values $2['x-'3']^2-'13'>37 \Rightarrow 2['x-'3']^2=50 \Rightarrow [x-3]^2=25 \Rightarrow x='3'\pm...$ Condone use of = rather than >
	A1	For both critical values of $x=-2,\ 8$
	M1	For choosing the outside region for their cv's [provided there are two values] $\{x<'-2'\}\cup\{x>'8'\}$ or any other correct notation. Condone 'AND' for this mark
	A1	For the correct region with correct values $\{x<-2\}\cup\{x>8\}$ or any other correct notation. Must **not** use 'AND' for this mark.
Alt – solving inequality without use of completed square form		
	M1	Sets f(x) > 37 and attempts to find **two** critical values. See general guidance on what constitutes an attempt to solve a quadratic. Condone use of = rather than >
	A1	For both critical values of $x=-2,\ 8$
	M1	For choosing the outside region for their cv's [provided there are two values] $\{x<'-2'\}\cup\{x>'8'\}$ or any other correct notation. Condone 'AND' for this mark
	A1	For the correct region with correct values $\{x<-2\}\cup\{x>8\}$ or any other correct notation. Must **not** use 'AND' for this mark.

Type MS · Completing the square

Answer 4 of 12 · Question 2 part a · 3 marks

2022 JAN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

	$(2x)^2 = (3x)^2 + (4x)^2 - 2\times 3x\times 4x\times \cos A$ **or** e.g. $(\cos A =)\dfrac{(3x)^2+(4x)^2-(2x)^2}{2\times 3x\times 4x} \Rightarrow \angle A = 28.95502...^\circ$ All the values here are correct to give 28.95502, but the labelling of angle A is incorrect. Allow angles in degrees to be rounded to the nearest whole number here and angles in radians to be rounded to 1 dp. Where candidates do not actually work the angle out, if and only if the fully correct expression for cos(angle) or angle = $\cos^{-1}$ is seen in the 1st M1, look for any of the following used in the 2nd M1, then this A mark can be awarded. . $\sin\left(\cos^{-1}(\text{ANS})\right)$ or $\sin\left(\cos^{-1}\left(((angle)\ A)\right)\right)$ or $\sin\left(\cos^{-1}\left(((angle)\ B)\right)\right)$ or $\sin\left(\cos^{-1}\left(((angle)\ C)\right)\right)$ or for $\sin\left(\cos^{-1}\left(-\frac{1}{4}\right)\right)$ or $\sin\left(\cos^{-1}\left(\frac{11}{16}\right)\right)$ or $\sin\left(\cos^{-1}\left(\frac{7}{8}\right)\right)$ seen
dM1	For using the correct formula for the area of a triangle using 50 and their angle (or use of sincos^{-1} correctly and the correct 2 sides for their angle and a fully correct rearrangement to find x^2 or x If 1st M1 has been gained and a poor rearrangement occurs to give an incorrect angle ft their angle. This is a general principle of marking. Sight of 17.2(1…….) following M1 A1 and the correct formula written will imply this mark. The same condition applies, if the angle is not actually worked out as for the A mark. Dependent on previous method mark.
A1	For the correct value of x. Accept awrt 4.15 [cm]
ALT1	
M1	For fully correct substitution, as shown into Heron's formula.
A1	For the correct expression for s
x must be present, but may be recovered (indicated by A1)	
dM1	Applies Heron's formula correctly with their value of s and 50 and a fully correct rearrangement to find x^2 or x
A1	For the correct value of x. Allow all awrt 4.15.

Type MS · Completing the square

Answer 5 of 12 · Question 2 part a · 3 marks

2024 JUN · Paper 01 · extract 2 of 4 · find part a in this mark scheme extract

ALT2	
M1	For a fully correct substitution into the cosine **formula** as shown. ie including an equals sign. OR for a fully correct **expression** for cosABC or cosBAC or cosACB Allow students to use just B, A and C for angles or any labelling.
A1	For one of the correct values for sin(relevant angle)
Note for M1 A1 The x can be consistently omitted **or** the x can be recovered (also indicated by a correct value for sin) **or** values in the correct proportions can be used as an alternative (also indicated by a correct value for sin). M1 may also be awarded if the values in the expression or equation lead to the correct sin value for those values, regardless of the labelling of the angle. This may also lead to A1 $(2x)^2 = (3x)^2 + (4x)^2 - 2\times 3x\times 4x\times \cos A$ **or** e.g. $(\cos A =)\dfrac{(3x)^2+(4x)^2-(2x)^2}{2\times 3x\times 4x} \Rightarrow \sin A = \sqrt{\dfrac{15}{64}}$ All the values here are correct to give $\Rightarrow (\sin ABC =)\sqrt{\dfrac{15}{64}}$, but the labelling of angle A is incorrect. Allow values to be rounded to 1 dp.	
dM1	For using the correct formula for the area of a triangle using 50 and their sin value correctly and the correct 2 sides for their angle and a fully correct rearrangement to find x^2 or x If M1 has been gained but their sin value is incorrect, students must be seen to have substituted into $\cos^2(\text{angle}) + \sin^2(\text{angle}) = 1$. Though a poor rearrangement to give an incorrect sin value, will potentially allow this mark to be awarded. Sight of 17.2(1…….) following M1 A1 will imply this mark. Dependent on previous method mark.
A1	For the correct value of x. Accept awrt 4.15 [cm]

Type MS · Completing the square

Answer 5 of 12 · Question 2 part a · 3 marks

2024 JUN · Paper 01 · extract 3 of 4 · find part a in this mark scheme extract

Question	Scheme	Marks
For part (a) of this question, mark using the scheme which gives the most marks.		
2 (a)	$A=2 \quad B=1 \quad C=7$	B1B1B1
ALT	$2x^2+4x+9=2\left(x^2+2x\right)+9=2\left[\left(x+1\right)^2-1\right]+9$ $\Rightarrow \mathrm{f}\left(x\right)=2\left(x+1\right)^2+7 \qquad A=2 \quad B=1 \quad C=7$	M1M1 A1 [3]
(b)(i)	-1	B1ft
(ii)	$\frac{1}{7}$	B1ft [2]
		Total 5 marks

Part	Mark	Notes
• **Mark using the B scheme first.** • If not full marks – use the MMA scheme also, if appropriate. • Same score – apply the B marks. • Higher score – apply the marks from the MMA scheme. Allow the candidate to state A, B and C or the values to be embedded within an expression.		
(a) Different marks on ePen	B1	For one of A, B, or C correct.
	B1	For two of A, B, or C correct.
	B1	For all of A, B, and C correct.
ALT	M1	For correctly factorising the given expression to achieve either: $2\left(x^2+2x\right)+9$ or $2\left(x^2+2x+\frac{9}{2}\right)$
	M1	Completes the square correctly, **regardless of any factor on the outside – follow through their factorisation.** $\left(x^2+ax+b\right)$ or $\left(x^2+ax\right)+c \Rightarrow$ ie $\left[\left(x+\frac{a}{2}\right)^2-\left(\frac{a}{2}\right)^2+b\right]$ or $\left[\left(x+\frac{a}{2}\right)^2-\left(\frac{a}{2}\right)^2\right]+c \qquad a,b,c\neq 0$
	A1	For all of A, B and C correct.
Note, the question does not ask students to show working nor preclude the use of a calculator, so if values are simply listed, these can be given marks from the B scheme throughout.		
(b)(i)	B1ft	For the correct value or follow through their $-B$
(ii)	B1ft	For the correct value or follow through their $\frac{1}{C}$
Only if no labelling of (i) and (ii)	SC2	If candidate clearly writes max value of $\frac{1}{7}$ when $x=-1$. Allow ft
	SC1	If candidate clearly writes $\left(-1,\frac{1}{7}\right)$ Allow ft Marked as 1st B1
If no labelling of (i) and (ii) is present for parts (b) – marks may be awarded for the values presented in the correct order. **B0 B0 if not labelled and work doesn't meet this condition.**		

Type MS · Completing the square

Answer 5 of 12 · Question 2 part a · 3 marks

2024 JUN · Paper 01 · extract 4 of 4 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Completing the square

Answer 6 of 12 · Question 2 part a · 3 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

M1	For either $11 = 2a+3$ **or** $b = a+2$
M1	For $11 = 2a+3$ **and** $b = a+2$
A1	$a = 4,\ \ b = 6$
ALT	
M1	Multiply by $2-\sqrt{3}$
M1	For either $a = 22-3b$ **or** $1 = 2b-11$
M1	For $a = 22-3b$ **and** $1 = 2b-11$
A1	$a = 4,\ \ b = 6$

Type MS · Completing the square

Answer 6 of 12 · Question 2 part a · 3 marks

2020 JAN · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks
2 (a)	$7+4x-x^2=11-(x-2)^2$ $[a=11, b=1, c=-2]$ **ALT** $7+4x-x^2=a-b(x^2+2cx+c^2)$ $a-bc^2=7 \quad b=1 \quad bc=4 \quad$ So $a=11, b=1, c=-2$ $7+4x-x^2=11-(x-2)^2$	M1A1A1 [3] {M1} {A1}{A1} [3]
(b)	(i) 11 (ii) 2	B1ft B1ft [2]
		Total 5 marks

Type MS · Completing the square

Answer 7 of 12 · Question 2 part a · 3 marks

2025 NOV · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

Question	Scheme	Marks
2 (a)	$f(x) = 2x^2 - 10x + 7 = 2\left(x^2 - 5x\right) + 7$	M1
	$= 2\left(x - \frac{5}{2}\right)^2 - \frac{11}{2}$	M1
	$\Rightarrow a = 2 \quad b = \frac{-5}{2} \quad c = \frac{-11}{2}$	A1 [3]
(b)	(i) $\frac{-11}{2}$	B1ft
	(ii) $\frac{5}{2}$	B1ft [2]
		Total 5 marks

Type MS · Completing the square

Answer 7 of 12 · Question 2 part a · 3 marks

2025 NOV · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Completing the square

Answer 8 of 12 · Question 2 part a · 4 marks

2021 JUN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

2 (a)	$2-\frac{1}{25}\left(x^2-20x\right)$	
	$2\mp\frac{1}{25}\left[\left(x\pm10\right)^2-100\right]$	M1
	$6-\frac{1}{25}\left(x-10\right)^2$ So $A=6$ $B=\frac{1}{25}$ $C=-10$	A1 A1 A1 (4)
ALT		
	$A-Bx^2-2BCx-BC^2=2+\frac{4}{5}x-\frac{1}{25}x^2$	{M1}
	$B=\frac{1}{25}$	{A1}
	$-\frac{2}{25}C=\frac{4}{5}$ $C=-10$	{A1}
	$-\frac{1}{25}(100)+A=2$ $A=6$	{A1} (4)
(b)(i)	6	B1 ft
(b)(ii)	10	B1 ft (2)
		Total 6 marks

Type MS · Completing the square

Answer 8 of 12 · Question 2 part a · 4 marks

2021 JUN · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Guidance
(a)	M1	For a complete method to complete the square to achieve as a minimum $2 \mp \frac{1}{25}(x \pm 10)^2 - p$ or $\mp \frac{1}{25}\left[(x \pm 10)^2 - q\right]$ where p and q are constants
	A1	For one correct from $A = 6$ $B = \frac{1}{25}$ or $C = -10$ whether stated explicitly or embedded
	A1	For two correct from $A = 6$ $B = \frac{1}{25}$ or $C = -10$ whether stated explicitly or embedded
	A1	Fully correct $A = 6$ $B = \frac{1}{25}$ and $C = -10$ **OR** $6 - \frac{1}{25}(x-10)^2$ oe
	ALT – equates coefficients	
	M1	For an attempt to expand $A - B(x+C)^2$ **AND** equate coefficients to the given $f(x) \Rightarrow A - Bx^2 - 2BCx - BC^2 = 2 + \frac{4}{5}x - \frac{1}{25}x^2$ Allow $A \pm Bx^2 \pm 2BCx \pm BC^2$ for the expansion of $A - B(x+C)^2$ There must be an attempt to equate at least one coefficient. $-B = -\frac{1}{25} \Rightarrow B = ...$ $-2BC = \frac{4}{5} \Rightarrow C = ...$ $A - BC^2 = 2 \Rightarrow A = ...$
	A1	For one correct from $A = 6$ $B = \frac{1}{25}$ or $C = -10$ whether stated explicitly or embedded
	A1	For two correct from $A = 6$ $B = \frac{1}{25}$ or $C = -10$ whether stated explicitly or embedded
	A1	Fully correct $A = 6$ $B = \frac{1}{25}$ and $C = -10$ **OR** $6 - \frac{1}{25}(x-10)^2$ oe
(b)(i)	B1ft	For the value of 6 or ft their A
(ii)	B1ft	For the value of 10 or ft their C

Type MS · Completing the square

Answer 8 of 12 · Question 2 part a · 4 marks

2021 JUN · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Completing the square

Answer 9 of 12 · Question 11 part a · 4 marks

2023 JUN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks
11(a)	$f(x)=10+6x-x^2=-(x^2-6x)+10 \Rightarrow A=-1$	B1
	$f(x)=-\left[(x-3)^2-9\right]+10=-(x-3)^2+19$	M1
	$B=-3,\ C=19$	A1A1
		[4]
(b)	(i) $x=3$	B1ft
	(ii) f $(x)_{\text{greatest}} = 19$	B1ft
		[2]
(c)	$x^2-x+13=10+6x-x^2 \Rightarrow 2x^2-7x+3=0$	M1
	$\Rightarrow (2x-1)(x-3)=0 \Rightarrow x=\frac{1}{2},\ 3$	M1A1
		[3]
(d)	$A=\int_{\frac{1}{2}}^{3}(10+6x-x^2)\,dx-\int_{\frac{1}{2}}^{3}(x^2-x+13)\,dx=\left[\int_{\frac{1}{2}}^{3}-2x^2+7x-3\,dx\right]$	M1
	$A=\left[-\frac{2x^3}{3}+\frac{7x^2}{2}-3x\right]_{\frac{1}{2}}^{3}$	M1A1
	$A=\left[\left(-\frac{2\times3^3}{3}+\frac{7\times3^2}{2}-3\times3\right)-\left(-\frac{2\times0.5^3}{3}+\frac{7\times0.5^2}{2}-3\times0.5\right)\right]$	M1
	$A=\frac{125}{24}$ oe	A1
		[5]
		Total 14 marks

Part	Mark	Notes
(a)	**B1**	For factorising – 1 and finding $A = -1$
	M1	For an attempt to complete the square – (See general guidance)
	A1	For either $B=-3$ or $C=19$
	A1	For both $B=-3$ and $C=19$ Accept all values embedded.
	NB: Correct values following no working – Award full marks in this part.	

Type MS · Completing the square

Answer 9 of 12 · Question 11 part a · 4 marks

2023 JUN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

(b)(i)	**B1ft**	$x = 3$ ft their B You may see differentiation to find a maximum x. Allow a correct value of x here even if it does not follow from their working.
(b)(ii)	**B1ft**	f $(x) = 19$ ft their C You may see differentiation to find a maximum x. Allow a correct value of f(x) here even if it does not follow from their working. Allow $y = 19$.
(c)	**M1**	Sets the equation of curve C equal to the equation of curve S and forms a 3TQ
	M1	Attempts to solve their 3TQ using any correct method to find **two** values of x If their 3TQ is incorrect and no working is seen to solve it, withhold this mark.
	A1	For both $x = \frac{1}{2}$, 3 correct Accept coordinates – ignore y values.
	NB: Both correct values following no working – Award full marks in this part.	
(d)	**M1**	For a correct statement with their values of x in the correct position of the intent to integrate the two expressions and **subtract** the result, or to integrate $2x^2 - 7x + 3$ Ft their limits for this mark, the correct way around, but allow recovery at a later stage. If they deal with the two expressions separately, look for subtraction at the end of their solution.
	M1	For an attempt to integrate their combined expression or **two** separate expressions. They must achieve as a minimum $\pm Px^3 \pm Qx^2 \pm Rx$ for their integration whether they integrate two expressions separately (each) or one combined expression. Ignore limits for this mark.
	A1	For the correct integrated expression(s)– ignore limits for this mark Allow $\frac{2x^3}{3} - \frac{7x^2}{2} + 3x$
	M1	For substituting their limits into their integrated expression and subtracting the results of lower limit from upper limit. **Note carefully:** If they have the correct limits and correct integration allow this mark if the **correct** area is obtained without seeing explicit substitution. $\left(\pm\frac{125}{24},\ \pm 5.208...\right)$ If their limits are incorrect or the integration is incorrect or the final area is incorrect then explicit substitution **must** be seen for the award of this mark.
	A1	For the correct area $A = \frac{125}{24}$ oe

Type MS · Completing the square

Answer 9 of 12 · Question 11 part a · 4 marks

2023 JUN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

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Type MS · Completing the square

Answer 10 of 12 · Question 10 part a · 4 marks

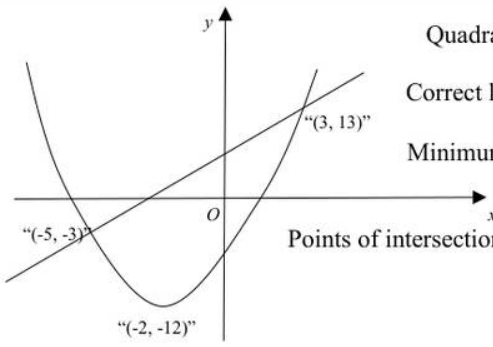
2024 NOV · Paper 01 · find part a in this mark scheme extract

Number		
10 (a)	$\left(3-9\left(\frac{4}{9}x+x^2\right)\right.$ or $\left.-9\left(-\frac{1}{3}+\frac{4}{9}x+x^2\right)\right)$ $3-9\left[\left(x+\frac{2}{9}\right)^2-\frac{4}{81}\right]$ or $-9\left[\left(x+\frac{2}{9}\right)^2-\frac{4}{81}-\frac{1}{3}\right]$ or $-1\left[9\left(x+\frac{2}{9}\right)^2-\frac{36}{81}-3\right]$ or $3+(1)\left[-9\left(x+\frac{2}{9}\right)^2+\frac{36}{81}\right]$ $\frac{31}{9}-9\left(x+\frac{2}{9}\right)^2$ $A=\frac{31}{9}$ $B=9$ $C=\frac{2}{9}$ oe	M1 A1 A1 A1 [4]
(b)	$"\frac{31}{9}"$	B1ft [1]
(c)	$\alpha+\beta=-\frac{4}{9}$ $\alpha\beta=-\frac{3}{9}$ oe $\frac{3\alpha}{\beta}+\frac{3\beta}{\alpha}\left(=\frac{3\left(\alpha^2+\beta^2\right)}{\alpha\beta}\right)=\frac{3\left((\alpha+\beta)^2-2\alpha\beta\right)}{\alpha\beta}$ $\frac{3\left[\left("-\frac{4}{9}"\right)^2-2\left("-\frac{3}{9}"\right)\right]}{"-\frac{3}{9}"}$ $\left(=-\frac{70}{9}\right)$ $\left(\frac{3\alpha}{\beta}\times\frac{3\beta}{\alpha}=\right)9$ $x^2+"\frac{70}{9}"x+"9"\,(=0)$ $9x^2+70x+81=0$	B1 M1 A1ft B1 M1 A1cso [6]
(d)	$\left((x+y)^3=\right)x^3+3x^2y+3xy^2+y^3=x^3+y^3+3xy(x+y)$	B1*cso [1]
(e)	$(\alpha^2-\beta+\beta^2-\alpha=)$ $(\alpha+\beta)^2-2\alpha\beta-(\alpha+\beta)$ $\left((\alpha+\beta)^2=(\alpha+\beta)^2-2\alpha\beta=\frac{70}{81}\right)$ $\left("-\frac{4}{9}"\right)^2-2\left("-\frac{3}{9}"\right)-\left("-\frac{4}{9}"\right)$ $\left(=\frac{106}{81}\right)$ **candidates may also transfer a value for $\alpha^2+\beta^2$ from working in part (c)** $\left[(\alpha^2-\beta)(\beta^2-\alpha)=\alpha^2\beta^2-\beta^3-\alpha^3+\alpha\beta\right]$ $=(\alpha\beta)^2+\alpha\beta-(\alpha^3+\beta^3)$ or $(\alpha\beta)^2+\alpha\beta-\alpha^3-\beta^3$ $\left[(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3=\alpha^3+\beta^3+3\alpha\beta(\alpha+\beta)\right]$ $=(\alpha\beta)^2+\alpha\beta-(\alpha+\beta)^3+3\alpha\beta(\alpha+\beta)$ or $(\alpha\beta)^2+\alpha\beta-\left((\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)\right)$ $\left[=\left("-\frac{3}{9}"\right)^2+\left("-\frac{3}{9}"\right)-\left("-\frac{4}{9}"\right)^3+3\left("-\frac{3}{9}"\right)\left("-\frac{4}{9}"\right)\left(=\frac{226}{729}\right)\right]$ $"\frac{106}{81}"=-\frac{q}{3}$ or $"\frac{226}{729}"=\frac{r}{3}$ $q=-\frac{106}{27}$ and $r=\frac{226}{243}$	M1 A1ft M1 M1 dddM1 A1 [6]
		Total 18 marks

Type MS · Completing the square

Answer 11 of 12 · Question 2 part a · 2 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

number		
2 (a)	$(x+2)^2-4-8$	M1
	$(x+2)^2-12$ $\quad a=2 \quad b=-12$	A1
		(2)
(b)	$x^2+4x-8=2x+7$	M1
	$x^2+2x-15=0$	dM1
	$(x-3)(x+5)=0$ or any valid method	M1
	$x=3, y=13$ $\quad x=-5, y=-3$	A1 A1
		(5)
(c)	 Quadratic drawn	B1
	Correct line drawn	B1
	Minimum labelled	B1 ft
	Points of intersection labelled	B1 ft
		(4)
		[11]

Part	Mark	Additional Guidance
(a)	M1	Use general guidance, allow an expression of the form $\left(x\pm\frac{4}{2}\right)^2\pm q\pm 8 \quad q\neq 0$
	A1	Correct expression as shown, a and b need not be explicitly stated
(b)	M1	Correctly equates the 2 expressions
	dM1	Rearranges to a 3TQ = 0 (allow any 3TQ if intention of rearrangement is clear)
	M1	Uses any valid method to solve – **see general guidance**
	A1	For either pair of values stated
	A1	For all four values, correctly paired or written as coordinates.
	For the final A1 A1, do not allow recovery of y values from part c.	
(c)	B1	Correctly shaped quadratic curve, with a clear minimum point, drawn anywhere on their axis, mark intention.
	B1	Correct line – **must have a positive y intercept, a positive gradient and a negative x intercept**
	B1ft	Correctly labelled coordinates for their minimum, ft their answer from b, **must correctly ft their answer from a, ie minimum point labelled (- *a*, *b*)**
	B1ft	Correctly labelled coordinates for their intersections.
	The coordinates must be clearly indicated and not inferred from a scale on the graph. Ignore any labelling of intersections with axes.	

Type MS · Completing the square

Answer 11 of 12 · Question 2 part a · 2 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Completing the square

Answer 12 of 12 · Question 2 part a · 3 marks

2023 NOV · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

	$\frac{\quad}{\left(2+4\sqrt{5}\right)} \times \frac{\quad}{\left(2-4\sqrt{5}\right)} = \frac{\quad}{-76}$ oe Allow one error in multiplying out. Look for equivalences. The question precludes the use of a calculator, so method must be shown.
A1	$4+3\sqrt{5}$ or $a=4,\ b=3$
ALT	
M1	For the correct statement for the area, this may be explicit or implied by working, but **must** be a fully correct statement. Allow $a+b\sqrt{5}$ to be written as BC or denoted by a letter.
M1	For multiplying out correctly as shown in the scheme. But if the initial formula is missing the $\frac{1}{2}$, the mark may also be awarded for $\left(34+11\sqrt{5}\right)=2a+2b\sqrt{5}+4a\sqrt{5}+20b$
M1	For correctly equating coefficients and any valid complete method to solve a pair of simultaneous equations, each equation in a and b, allow one error. This work may also be done with the $\frac{1}{2}$ missing from the formula and that doesn't count as 'one error'. The question precludes the use of a calculator, so method must be shown.
A1	$4+3\sqrt{5}$ or $a=4,\ b=3$

Type MS · Completing the square

Answer 12 of 12 · Question 2 part a · 3 marks

2023 NOV · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks
For part (a) of this question, mark using the scheme which gives the most marks.		
2a	$p=2,\quad q=\frac{1}{8},\quad r=-\frac{97}{32}$	B1B1B1
ALT	$\left(g(x)=2x^2+\frac{1}{2}x-3=\right)2\left(x^2+\frac{1}{4}x-\frac{3}{2}\right)$ or $2\left(x^2+\frac{1}{4}x\right)-3$	M1
	$("2")\left(\left[x+\frac{1}{8}\right]^2-\frac{1}{64}-\frac{3}{2}\right)$ or $("2")\left(\left[x+\frac{1}{8}\right]^2-\frac{1}{64}\right)-3$	M1
	$2\left[x+\frac{1}{8}\right]^2-\frac{97}{32}$ or $p=2,\quad q=\frac{1}{8},$ oe $r=-\frac{97}{32}$ oe	A1 [3]
(b)	(i) $\left(g(x)_{\min}=\right)"-\frac{97}{32}"$	B1ft
	(ii) $\left(x=\right)-"\frac{1}{8}"$	B1ft [2]
(c)	(i) $h(x)_{\min}="-\frac{97}{32}"$	B1ft
	(ii) $x=\sqrt[3]{"-\frac{1}{8}"}=-\frac{1}{2}$	B1ft [2]
		Total 7 marks

Part	Mark	Notes
• **Mark using the B scheme first.** • If not full marks – use the MMA scheme also, if appropriate. • Same score – apply the B marks. • Higher score – apply the marks from the MMA scheme.		
(a) **Different to Epen marks**	B1	For one of p, q, or r correct.
	B1	For two of p, q, or r correct.
	B1	For all of p, q, or r correct – note q and r can be written as equivalent fractions.
ALT	M1	Correctly takes out 2 as a common factor to give either of the 2 expressions shown.
	M1	Completes the square correctly, **regardless of any factor on the outside – follow through their factorisation.** $\left(x^2+ax+b\right)$ or $\left(x^2+ax\right)+c\Rightarrow$ ie $\left[\left(x+\frac{a}{2}\right)^2-\left(\frac{a}{2}\right)^2+b\right]$ or $\left[\left(x+\frac{a}{2}\right)^2-\left(\frac{a}{2}\right)^2\right]+c\qquad a,b,c\neq 0$
	A1	Fully correct answer – note q and r can be written as equivalent fractions.
Note part (a) does not ask students to show working nor preclude the use of a calculator, so if values are simply listed, these can be given marks from the B scheme		
(b) (i)	B1ft	For the correct value **or** ft their r
(ii)	B1ft	For the correct value **or** ft their $-q$
If no labelling of (i) and (ii) is present for parts (b) and (c) – marks may be awarded for the values presented in the correct order. **B0 B0 if not labelled and work doesn't meet this condition.**		
(c) (i)	B1ft	For the correct value **or** ft their $-\frac{97}{32}$
(ii)	B1ft	For the correct value **or** ft the cube root of their $-\frac{1}{8}$ (evaluated).

Type MS · Completing the square

Answer 12 of 12 · Question 2 part a · 3 marks

2023 NOV · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks